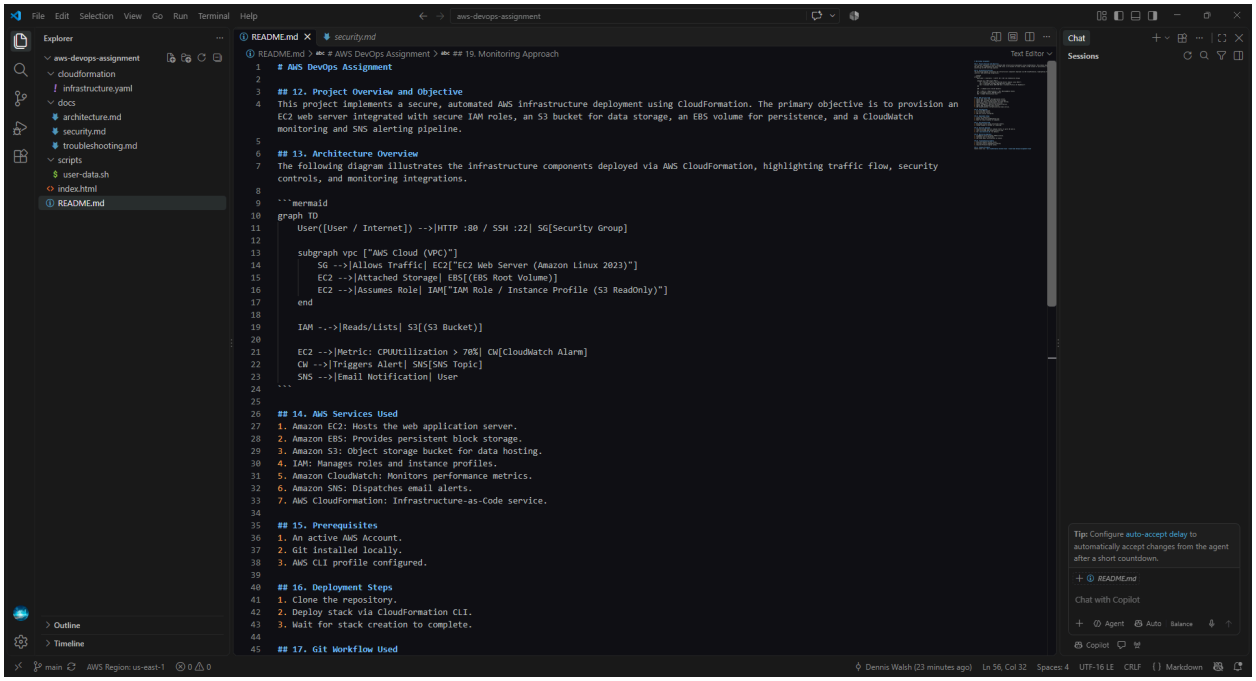


```
# AWS DevOps Assignment

## 12. Project Overview and Objective
This project implements a secure, automated AWS infrastructure deployment using CloudFormation. The primary objective is to provision an EC2 web server integrated with secure IAM roles, an S3 bucket for data storage, an EBS volume for persistence, and a CloudWatch monitoring and SNS alerting pipeline.

## 13. Architecture Overview
The following diagram illustrates the infrastructure components deployed via AWS CloudFormation, highlighting traffic flow, security controls, and monitoring integrations.

```mermaid
graph TD
    User([User / Internet]) -->|HTTP :80 / SSH :22| SG[Security Group]

    subgraph vpc ["AWS Cloud (VPC)"]
        SG -->|Allows Traffic| EC2["EC2 Web Server (Amazon Linux 2023)"]
        EC2 -->|Attached Storage| EBS[(EBS Root Volume)]
        EC2 -->|Assumes Role| IAM["IAM Role / Instance Profile (S3 ReadOnly)"]
    end

    IAM -.->|Reads/Lists| S3[(S3 Bucket)]

    EC2 -->|Metric: CPUUtilization > 70%| CW[CloudWatch Alarm]
    CW -->|Triggers Alert| SNS[SNS Topic]
    SNS -->|Email Notification| User
```

14. AWS Services Used
1. Amazon EC2: Hosts the web application server.
2. Amazon EBS: Provides persistent block storage.
3. Amazon S3: Object storage bucket for data hosting.
4. IAM: Manages roles and instance profiles.
5. Amazon CloudWatch: Monitors performance metrics.
6. Amazon SNS: Dispatches email alerts.
7. AWS CloudFormation: Infrastructure-as-Code service.

15. Prerequisites
1. An active AWS Account.
2. Git installed locally.
3. AWS CLI profile configured.

16. Deployment Steps
1. Clone the repository.
2. Deploy stack via CloudFormation CLI.
3. Wait for stack creation to complete.

17. Git Workflow Used
```

